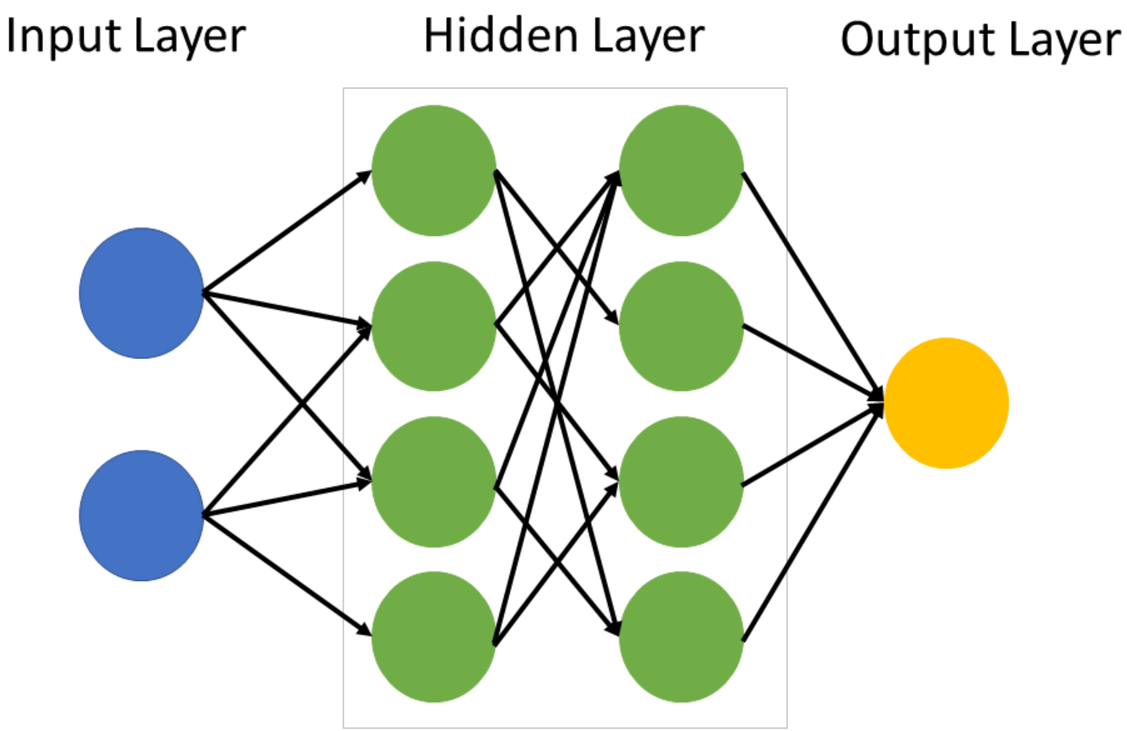
MATH50003 Numerical Analysis

VII Future Directions

Dr Sheehan Olver

Machine Learning (ML)

Numerics and automatic-differentiation underpins training



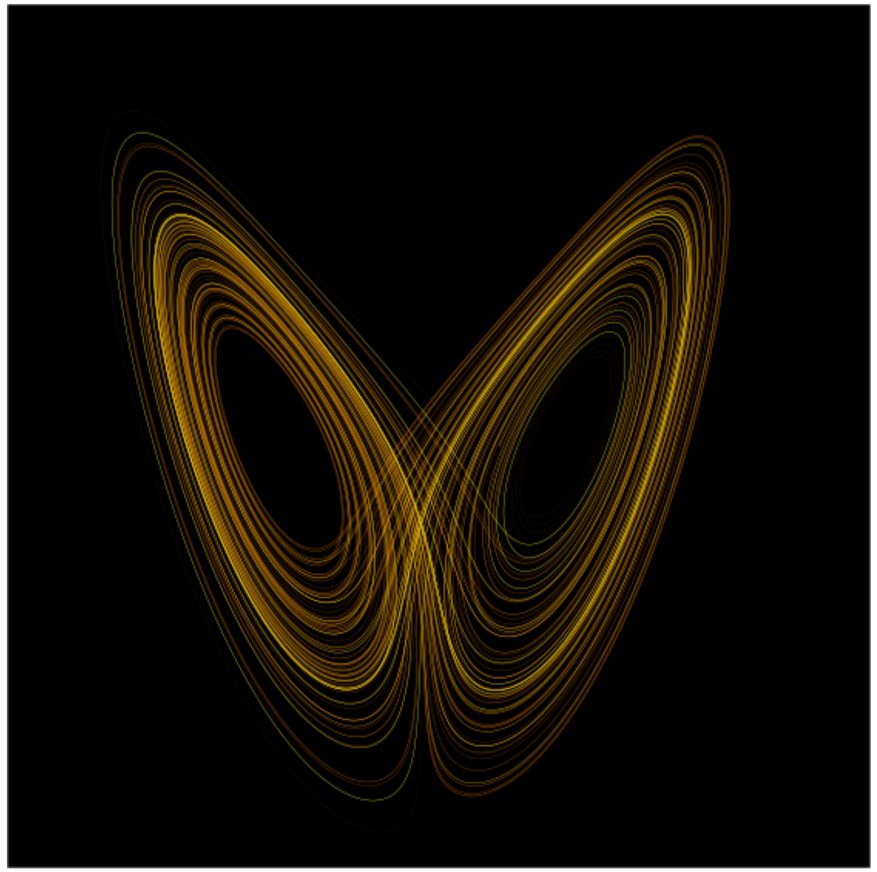
Artificial Neural Networks

MATH70005 Optimisation
MATH70134 Foundations of ML

Image from Wikipedia

Solving ordinary differential equations

Time-stepping beyond Euler's methods



$$\frac{\mathrm{d}x}{\mathrm{d}t} = \sigma(y - x),$$

$$\frac{\mathrm{d}y}{\mathrm{d}t} = x(\rho - z) - y,$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = xy - \beta z.$$

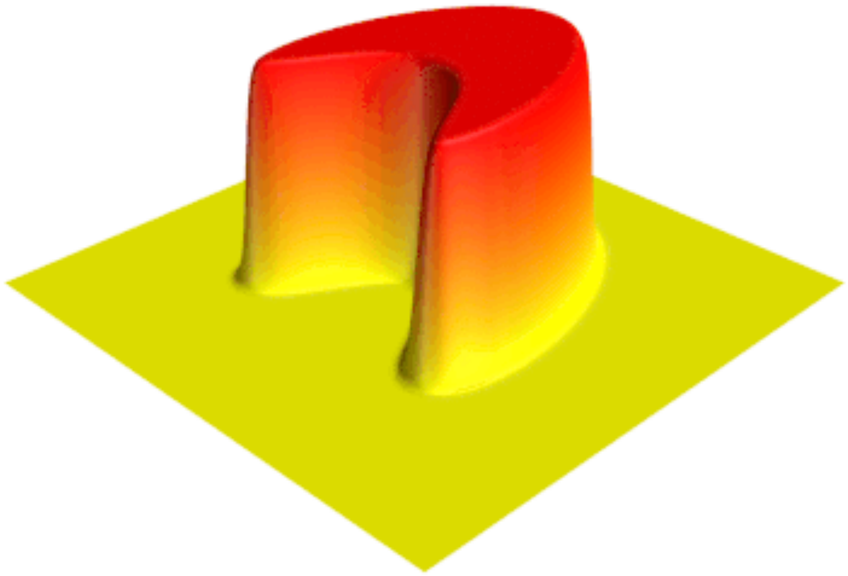
MATH70023 Numerical ODEs
MATH70027 Sci. Comput.

Image from Wikipedia

Solving partial differential equations

Finite Element Methods and Spectral Methods

$$u_t = u_{xx} + u_{yy}$$

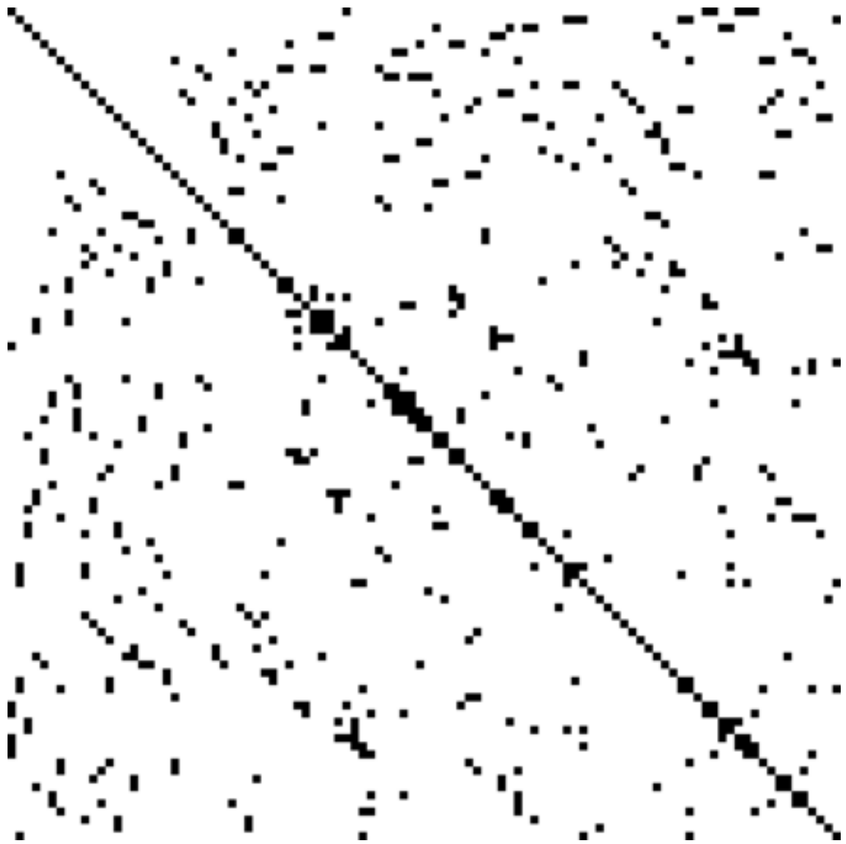


MATH70022 Finite Elements
MATH70025 Computational PDEs

Image from Wikipedia

Computational Linear Algebra

Solving linear systems coming from PDEs using iterative methods



MATH70024 Comput. Lin. Alg.

Image from Wikipedia

Scientific ML (SciML)

Combining differential equations and Neural Networks

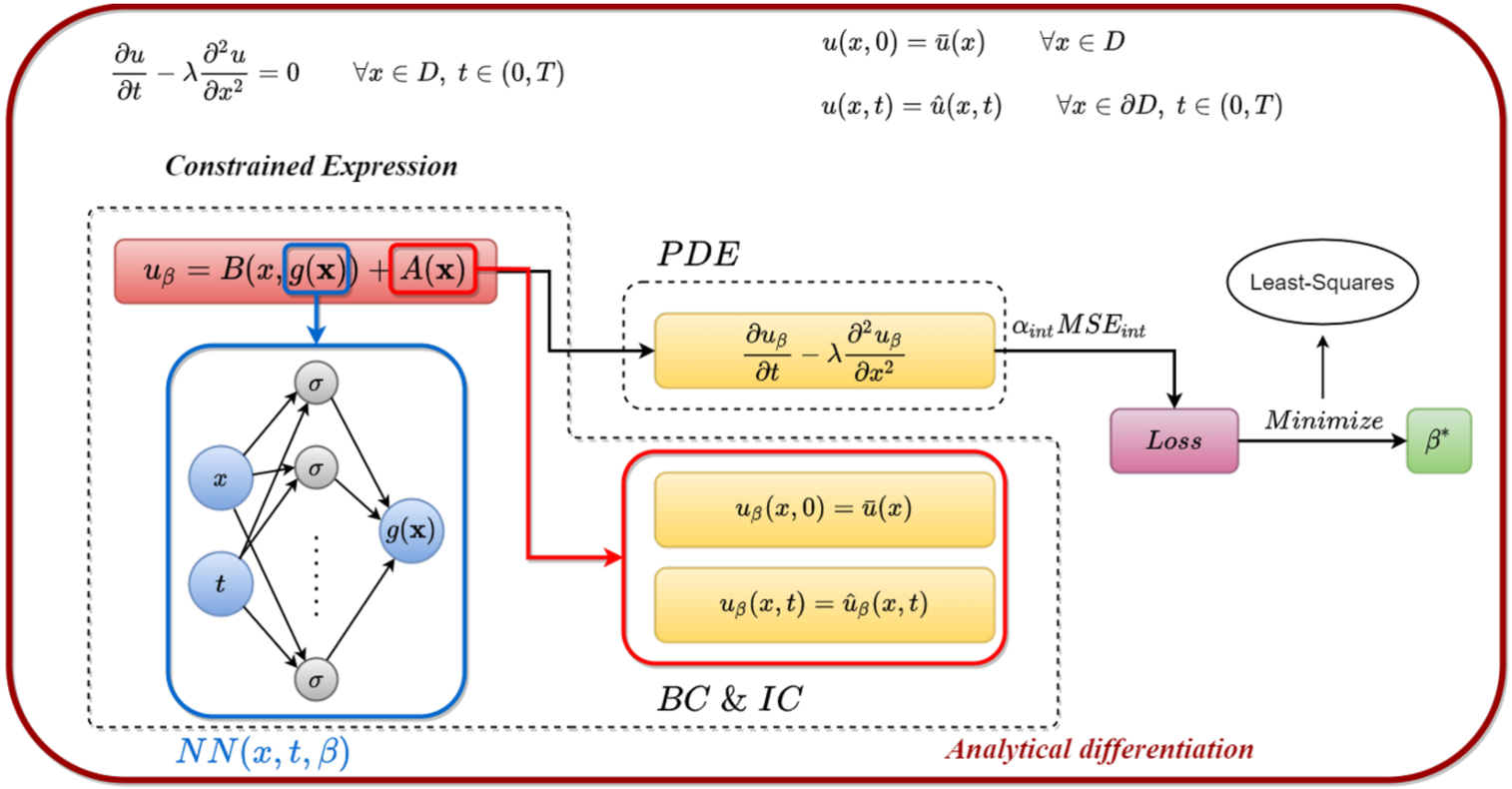


Image from Wikipedia

Non-examinable content (May 2026 only)

Final will be same for resitting students from last year

- II.1 Reals
- III.1 Structured Matrices
- III.2 LU & PLU factorisation
- III.5.1 Reduced QR and Gram–Schmidt
- IV.2 Singular Value Decomposition
- V.3 Discrete Convolutions
- VI.1.3 Stability of Chebyshev interpolation